usb_power
File: usb_power.kicad_sch

USB-C sink (no host), VBUS protection, USB0 device/FEL

power
File: power.kicad_sch

Regulators, rail order, supervisors, staged-power links

f133
File: f133.kicad_sch

F133-B SoC: supplies, clock, reset, pin assignment

boot_nor
File: boot_nor.kicad_sch

32 MiB SPI NOR boot (BOOT-SEL 00), straps, FEL/clip recovery

mn820
File: mn820.kicad_sch

MN820 (BT820B + 128 MiB DDR3L) module socket, host SPI1, LVDS to R09 panel

fpga
File: fpga_ecp5.kicad_sch

LFE5U-45F-6BG381C (optional accelerator): rails, sysCONFIG slave SPI, JTAG, SPI0 isolation

psram
File: psram.kicad_sch

APS6404L 8 MiB PSRAM: switched VDD (DONE-gated), static pulls

sd_audio
File: sd_audio.kicad_sch

microSD game media (not boot), 2x MAX98357A I2S speakers

controls
File: controls.kicad_sch

Game buttons, LEDs, debug UART

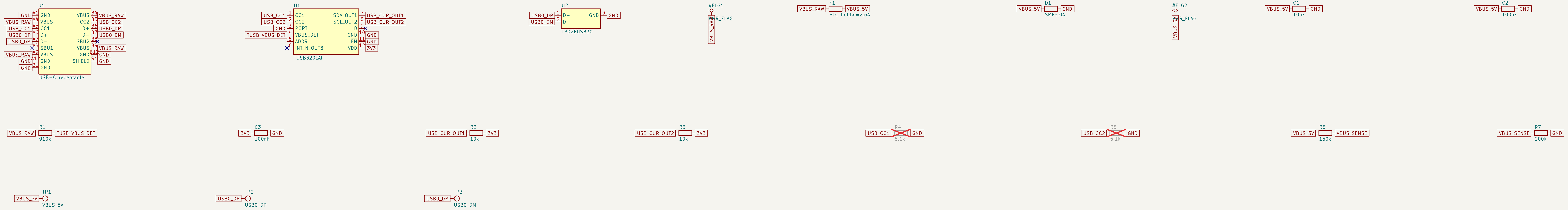
NOT approved for fabrication. LCD/panel/backlight deferred to R09.

Sheet: /
File: ARC1_R85.kicad_sch

Title: ARC-1 Compact R8.5 — hierarchical root

Size: A3
Date: 2026-10-15
Rev: R85-draft
KiCad E.D.A. 9.0.9
Id: 1/10

USB-C sink (no host), VBUS protection, USB0 device/FEL
Sheet generated from design.py; global labels carry every net.



Generated by R85_WORK/cad/gen/gen_kicad.py from design.py — edit the source, not this file
SELECTED_UNVERIFIED engineering capture — NOT released for fabrication
STRUTHIO ARC−1
Sheet: /usb_power/
File: usb_power.kicad_sch
Title: ARC−1 Compact R8.5: USB−C sink (no host), VBUS protection, USB0 d
Size: User
Date: 2026−10−15
Rev: R85−draft
KiCad E.D.A. 9.0.9
Id: 2/10

Regulators, rail order, supervisors, staged—power links
Sheet generated from design.py; global labels carry every net.

U3: 1 EN VBUS_5V; 2 GND GND; 3 SW SW_3V3; 4 VIN VBUS_5V; 5 PG PG_3V3_BUCK; 6 FB FB_3V3; 3V3 2A
U4: 1 RESET PG_3V3; 2 GND GND; 3 MR SUP33_MR_N; 4 CT SUP33_CT; 5 SENSE 3V3; 6 VDD 3V3; TPS3808G33
U5: 1 EN PG_3V3; 2 GND GND; 3 SW SW_CORE; 4 VIN VBUS_5V; 5 FB FB_CORE; VDD_CORE 0.95V
U6: 1 EN PG_3V3; 2 GND GND; 3 SW SW_DRAM; 4 VIN VBUS_5V; 5 FB FB_DRAM; 1V8_DRAM
U7: 1 IN 3V3; 2 GND GND; 3 EN 3V3; 4 NC; 5 OUT 1V8_AN; 1V8_AN
U8: 1 RESET F133_RESET_N; 2 GND GND; 3 MR RST_BTN_N; 4 CT SUPC_CT; 5 SENSE VDD_CORE; 6 VDD 3V3; TPS3808G09
U9: 1 EN FPGA_PWR_EN; 2 GND GND; 3 SW SW_1V1; 4 VIN VBUS_5V; 5 PG FPGA_VCC_PG; 6 FB FB_1V1; 1V1_FPGA
U10: 1 VIN 3V3; 2 GND GND; 3 ON FPGA_PWR_EN; 4 CT SW_FPGA_CT; 5 QOD 3V3_FPGA; 6 VOUT 3V3_FPGA; 3V3_FPGA switch
U30: 1 IN 3V3_FPGA; 2 GND GND; 3 EN 3V3_FPGA; 4 NC; 5 OUT 2V5_AUX_REG; 2V5_FPGA_AUX
U11: 1 VIN 3V3; 2 GND GND; 3 ON MN_PWR_EN; 4 CT SW_MN_CT; 5 QOD 3V3_MN; 6 VOUT 3V3_MN; 3V3_MN switch
U12: 1 VIN 3V3_FPGA; 2 GND GND; 3 ON PSRAM_EN; 4 CT SW_PS_CT; 5 QOD PSRAM_VDD; 6 VOUT PSRAM_VDD; PSRAM_VDD switch
U13: 1 A PSRAM_PWR_REQ; 2 B DONE_DLY; 3 GND GND; 4 Y PSRAM_EN; 5 VCC 3V3; AND
U14: 1 RESET PSRAM_PG; 2 GND GND; 3 MR PSRAM_SUP_MR_N; 4 CT PSRAM_SUP_CT; 5 SENSE PSRAM_VDD; 6 VDD 3V3_FPGA; TPS3808G30
J11: 1 3V3; 2 FPGA_PWR_EN; 3 GND; FPGA power override
SW1: 1 RST_BTN_N; 2 GND; RESET
C4 VBUS_5V GND 10uF
L1 SW_3V3 3V3_REG 2.2uH
R8 3V3_REG FB_3V3 453k
R9 FB_3V3 GND 100k
C5 3V3_REG GND 22uF
C6 3V3_REG GND 22uF
R10 PG_3V3_BUCK 3V3_REG 100k
#FLG3 PWR_FLAG 3V3_REG
R11 3V3_REG 3V3 0R
#FLG4 PWR_FLAG 3V3
TP4 3V3_REG
TP5 3V3
TP6 PG_3V3_BUCK
C7 SUP33_CT GND 2.2nF
R12 SUP33_MR_N 3V3 100k
R13 PG_3V3 3V3 10k
C8 3V3 GND 100nF
C9 VBUS_5V GND 10uF
L2 SW_CORE VDD_CORE_REG 2.2uH
R14 VDD_CORE_REG FB_CORE 57.6k
R15 FB_CORE GND 100k
C10 VDD_CORE_REG GND 22uF
C11 VDD_CORE_REG GND 22uF
#FLG5 PWR_FLAG VDD_CORE_REG
R16 VDD_CORE_REG VDD_CORE 0R
#FLG6 PWR_FLAG VDD_CORE
TP7 VDD_CORE
C12 VBUS_5V GND 10uF
L3 SW_DRAM 1V8_DRAM 2.2uH
R17 1V8_DRAM FB_DRAM 200k
R18 FB_DRAM GND 100k
C13 1V8_DRAM GND 22uF
#FLG7 PWR_FLAG 1V8_DRAM
TP8 1V8_DRAM
C14 3V3 GND 1uF
C15 1V8_AN GND 1uF
TP9 1V8_AN
C16 SUPC_CT GND 15nF
R19 RST_BTN_N 3V3 100k
R20 F133_RESET_N 1V8_LDOA 10k
TP10 F133_RESET_N
TP11 PG_3V3
C17 VBUS_5V GND 10uF
L4 SW_1V1 1V1_FPGA_REG 2.2uH
R21 1V1_FPGA_REG FB_1V1 82.5k
R22 FB_1V1 GND 100k
C18 1V1_FPGA_REG GND 22uF
C19 1V1_FPGA_REG GND 22uF
#FLG8 PWR_FLAG 1V1_FPGA_REG
R23 1V1_FPGA_REG 1V1_FPGA 0R
#FLG9 PWR_FLAG 1V1_FPGA
R24 FPGA_PWR_EN_F FPGA_PWR_EN 1k
R25 FPGA_PWR_EN GND 100k
R26 FPGA_VCC_PG 3V3 100k
TP12 1V1_FPGA
TP13 FPGA_VCC_PG
TP14 FPGA_PWR_EN
C20 SW_FPGA_CT GND 1nF
C21 3V3 GND 1uF
C22 3V3_FPGA GND 10uF
TP15 3V3_FPGA
C23 3V3_FPGA GND 1uF
C24 2V5_AUX_REG GND 1uF
FB2 2V5_AUX_REG 2V5_FPGA_AUX ferrite 120R@100MHz
#FLG10 PWR_FLAG 2V5_FPGA_AUX
TP16 2V5_FPGA_AUX
C25 SW_MN_CT GND 1nF
C26 3V3 GND 1uF
C27 3V3_MN GND 22uF
R27 MN_PWR_EN 3V3 100k
TP17 3V3_MN
C28 SW_PS_CT GND 1nF
C29 3V3_FPGA GND 1uF
C30 PSRAM_VDD GND 4.7uF
C31 3V3 GND 100nF
R28 PSRAM_PWR_REQ GND 100k
R29 PSRAM_EN GND 100k
C32 PSRAM_SUP_CT GND 1nF
R30 PSRAM_SUP_MR_N 3V3_FPGA 100k
R31 PSRAM_PG 3V3_FPGA 10k
C33 3V3_FPGA GND 100nF
TP18 PSRAM_VDD
TP19 PSRAM_PG
TP20 PSRAM_EN
TP21 GND
TP22 GND
TP23 GND
TP24 GND
#FLG11 PWR_FLAG GND
FID1 fiducial
FID2 fiducial
FID3 fiducial
H1 GND M2.5
H2 GND M2.5
H3 GND M2.5
H4 GND M2.5

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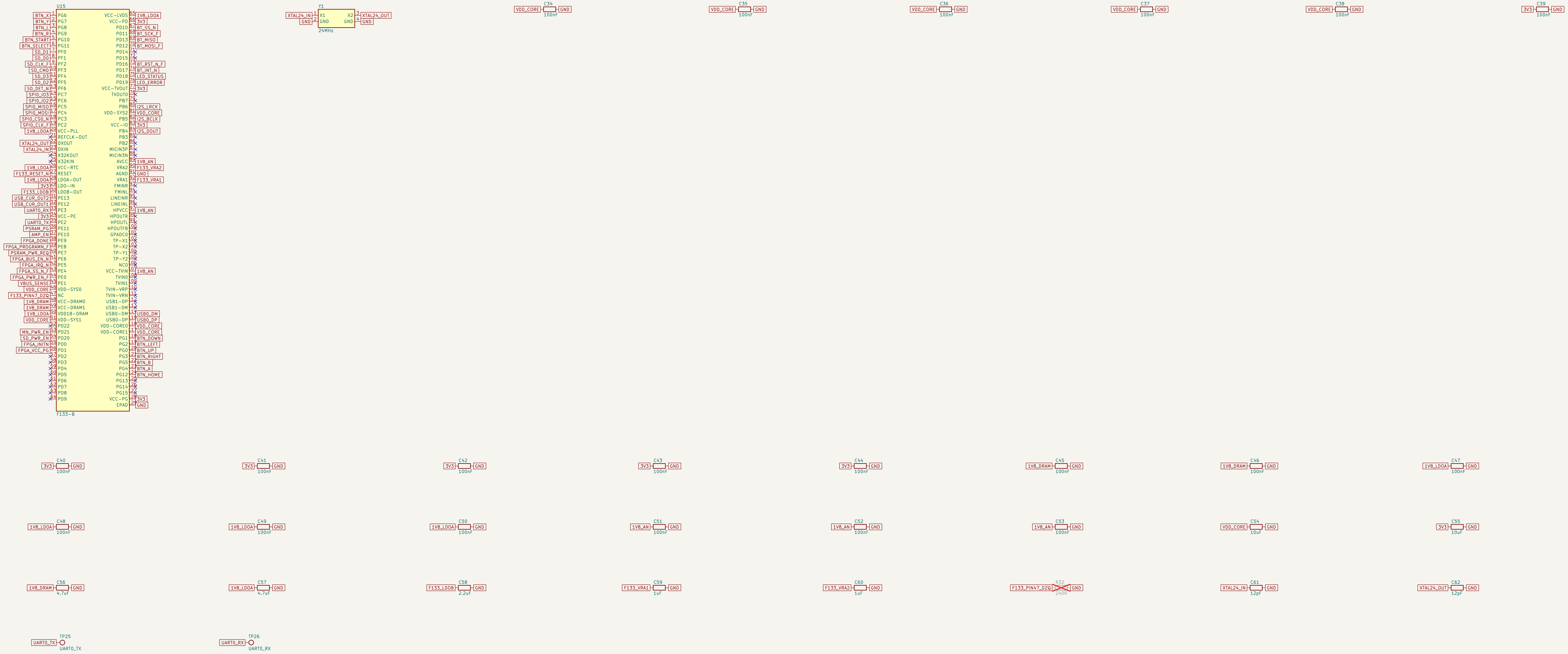
SELECTED_UNVERIFIED engineering capture — NOT released for fabrication
STRUTHIO ARC—1
Sheet: /power/
File: power.kicad_sch

Title: ARC—1 Compact R8.5: Regulators, rail order, supervisors, staged—powe

Size: User	Date: 2026—10—15	**Rev: R85—draft**
KiCad E.D.A. 9.0.9		Id: 3/10

F133-B SoC: supplies, clock, reset, pin assignment
Sheet generated from design.py; global labels carry every net.



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SELECTED_UNVERIFIED engineering capture — NOT released for fabrication

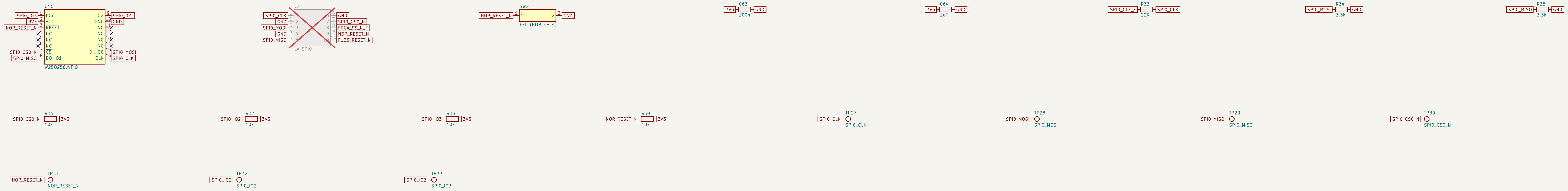
STRUTHIO ARC-1

Sheet: /f133/
File: f133.kicad_sch

Title: ARC-1 Compact R8.5: F133-B SoC: supplies, clock, reset, pin assignm

Size: User	Date: 2026-10-15	**Rev: R85-draft**
KiCad E.D.A. 9.0.9		Id: 4/10

32 MiB SPI NOR boot (BOOT-SEL 00), straps, FEL/clip recovery
Sheet generated from design.py; global labels carry every net.



Generated by R85_WORK/cad/gen/gen_kicad.py from design.py — edit the source, not this file
SELECTED_UNVERIFIED engineering capture — NOT released for fabrication
STRUTHIO ARC-1
Sheet: /boot_nor/
File: boot_nor.kicad_sch
Title: ARC-1 Compact R8.5: 32 MiB SPI NOR boot (BOOT-SEL 00), straps, F
Size: User | Date: 2026-10-15 | **Rev: R85-draft**
KiCad E.D.A. 9.0.9 | Id: 5/10

MN820 (BT820B + 128 MiB DDR3L) module socket, host SPI1, LVDS to R09 panel
Sheet generated from design.py; global labels carry every net.

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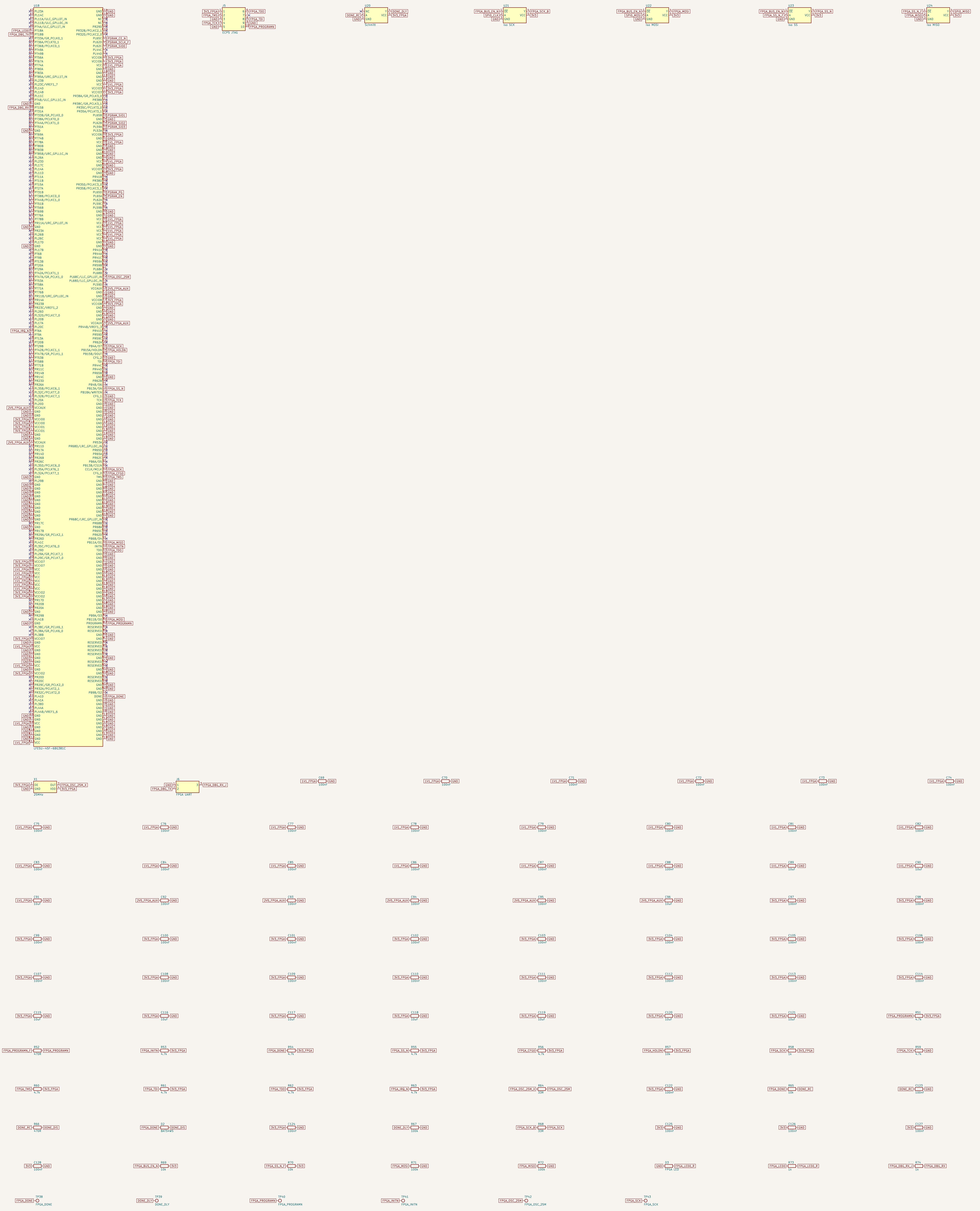
SELECTED_UNVERIFIED engineering capture — NOT released for fabrication
STRUTHIO ARC-1
Sheet: /mn820/
File: mn820.kicad_sch
Title: ARC-1 Compact R8.5: MN820 (BT820B + 128 MiB DDR3L) module sock
Size: User
Date: 2026-10-15
Rev: R85-draft
KiCad E.D.A. 9.0.9
Id: 6/10

LFE5U−45F−6BG381C (optional accelerator): rails, sysCONFIG slave SPI, JTAG, SPI0 isolation
Sheet generated from design.py; global labels carry every net.



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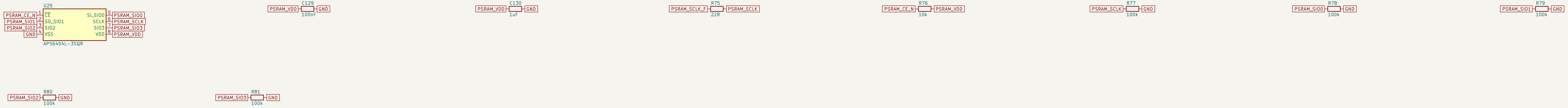
SELECTED_UNVERIFIED engineering capture − NOT released for fabrication
STRUTHIO ARC−1

Sheet: /fpga/
File: fpga_ecp5.kicad_sch

Title: ARC−1 Compact R8.5: LFE5U−45F−6BG381C (optional accelerator): rai

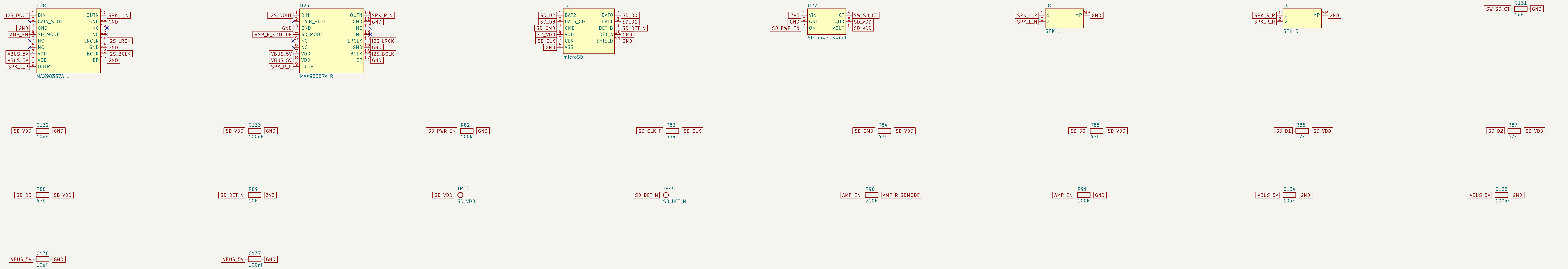
Size: User	Date: 2026−10−15	**Rev: R85−draft**
KiCad E.D.A. 9.0.9		Id: 7/10

APS6404L 8 MiB PSRAM: switched VDD (DONE–gated), static pulls
Sheet generated from design.py; global labels carry every net.



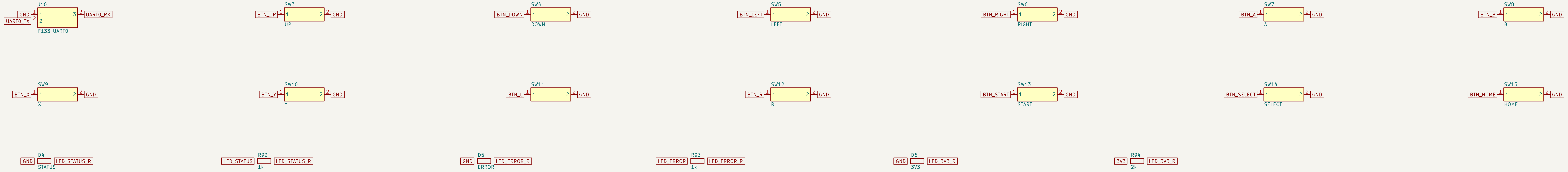
Generated by R85_WORK/cad/gen/gen_kicad.py from design.py – edit the source, not this file
SELECTED_UNVERIFIED engineering capture – NOT released for fabrication
STRUTHIO ARC–1
Sheet: /psram/
File: psram.kicad_sch
Title: ARC–1 Compact R8.5: APS6404L 8 MiB PSRAM: switched VDD (DONE–g
Size: User
Date: 2026–10–15
Rev: R85–draft
KiCad E.D.A. 9.0.9
Id: 8/10

microSD game media (not boot), 2x MAX98357A I2S speakers
Sheet generated from design.py; global labels carry every net.



Generated by R85_WORK/cad/gen/gen_kicad.py from design.py — edit the source, not this file
SELECTED_UNVERIFIED engineering capture — NOT released for fabrication
STRUTHIO ARC-1
Sheet: /sd_audio/
File: sd_audio.kicad_sch
Title: ARC-1 Compact R8.5: microSD game media (not boot), 2x MAX98357A
Size: User
Date: 2026-10-15
Rev: R85-draft
KiCad E.D.A. 9.0.9
Id: 9/10

Game buttons, LEDs, debug UART
Sheet generated from design.py; global labels carry every net.



Generated by R85_WORK/cad/gen/gen_kicad.py from design.py — edit the source, not this file
SELECTED_UNVERIFIED engineering capture — NOT released for fabrication
STRUTHIO ARC—1
Sheet: /controls/
File: controls.kicad_sch
Title: ARC—1 Compact R8.5: Game buttons, LEDs, debug UART
Size: User
Date: 2026—10—15
Rev: R85—draft
KiCad E.D.A. 9.0.9
Id: 10/10